

MELBOURNE MOORABBIN, AustraliaWMO: **948700**Lat: **37.9800S**Lon: **145.0964E**Elev: **13**StdP: **101.17**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.7	4.0	0.7	4.0	10.4	1.6	4.3	9.7	13.2	13.5	12.2	13.5	1.2	0	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	34.6	19.3	31.9	18.6	29.3	18.3	21.5	28.3	20.6	27.2	19.7	26.2	6.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.5	22.9	18.6	13.5	22.6	17.6	12.6	21.9	62.6	28.2	59.3	27.1	56.4	26.2	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	10.4	9.3	DB	0.2	39.8	1.1	2.6	-0.6	41.7	-1.2	43.2	-1.8	44.7	-2.6	46.5
			WB	-0.1	23.3	1.1	1.3	-0.9	24.2	-1.5	24.9	-2.2	25.7	-2.9	26.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	19.9	20.1	18.3	15.1	12.8	10.5	10.1	10.8	12.3	14.2	16.2	18.0
	DBStd	4.60	3.84	3.51	3.38	2.84	2.41	2.06	1.98	2.17	2.69	3.14	3.48	3.65
	HDD10.0	66	0	0	0	1	3	18	23	15	5	1	0	0
	HDD18.3	1541	23	18	43	103	173	236	256	234	183	136	85	50
	CDD10.0	1826	308	282	256	155	89	32	26	39	74	130	186	249
	CDD18.3	259	72	66	41	8	1	0	0	0	2	7	21	41
	CDH23.3	2784	763	624	402	76	2	0	0	0	9	126	275	507
	CDH26.7	1214	382	277	158	15	0	0	0	0	1	35	104	243
Wind	WSAvg	4.6	4.8	4.6	4.4	3.9	4.2	4.1	4.7	4.8	4.8	4.8	4.7	4.8
Precipitation	PrecAvg	724	49	41	49	64	76	57	65	72	67	72	60	58
	PrecMax	966	137	210	130	158	160	140	105	128	121	165	181	125
	PrecMin	444	1	1	12	8	23	15	17	21	25	22	5	3
	PrecStd	128	32	47	29	34	34	28	22	24	26	34	39	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	36.6	35.1	28.8	22.6	18.2	17.6	20.8	24.6	30.6	33.0	37.0
		MCWB	20.8	19.7	19.4	16.7	14.4	13.0	11.8	13.5	14.7	16.6	18.2	19.1
	2%	DB	34.6	33.2	30.7	25.4	20.3	16.5	15.9	18.2	21.8	26.7	29.8	32.7
		MCWB	19.5	19.4	18.5	16.0	14.0	12.4	11.3	12.0	14.0	15.9	17.7	18.3
	5%	DB	31.0	30.3	27.7	23.0	18.7	15.2	14.5	16.4	19.7	23.8	26.5	28.9
		MCWB	19.4	19.1	17.8	15.5	13.4	11.6	10.5	11.3	13.1	14.9	16.8	17.7
	10%	DB	27.2	27.1	24.8	20.8	17.2	14.1	13.6	14.9	17.7	20.9	23.3	25.3
		MCWB	19.1	18.9	17.6	15.0	12.9	11.1	10.0	10.5	12.3	13.9	16.2	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	22.6	21.2	18.3	16.2	14.2	13.1	14.2	16.3	18.2	20.5	21.3
		MCDB	30.0	28.2	28.2	24.6	19.4	16.2	16.1	18.8	21.4	25.4	27.9	28.3
	2%	WB	21.7	21.3	20.0	17.2	15.0	13.1	11.9	12.7	14.8	16.8	19.0	19.9
		MCDB	29.1	27.2	26.3	21.8	18.4	15.4	14.6	17.1	20.5	23.9	26.1	27.4
	5%	WB	20.6	20.5	19.0	16.4	14.1	12.2	11.1	11.7	13.6	15.7	18.0	18.9
		MCDB	27.6	26.5	25.2	21.1	17.6	14.5	13.7	15.3	18.6	22.0	24.4	26.1
	10%	WB	19.6	19.6	18.1	15.5	13.3	11.5	10.5	10.9	12.6	14.4	16.8	17.9
		MCDB	26.1	25.5	23.7	20.0	16.5	13.8	12.9	14.1	16.7	19.9	22.3	24.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.1	9.6	8.8	7.1	6.6	6.3	6.9	8.0	9.3	9.6	10.1
		MCDBR	18.6	17.0	15.6	13.5	9.6	8.3	7.8	9.6	11.6	16.0	16.8	18.7
		MCWBR	5.9	5.6	5.7	6.0	5.1	5.2	4.7	5.1	5.7	6.9	6.6	6.4
	5% WB	MCDBR	13.8	12.4	11.8	10.8	8.3	6.9	6.6	8.5	10.2	13.4	13.7	13.9
		MCWBR	5.7	5.5	5.3	5.5	5.0	4.7	4.4	5.0	5.6	6.7	6.2	6.2
Clear-Sky Solar Irradiance	taub	0.355	0.359	0.335	0.332	0.316	0.306	0.307	0.318	0.341	0.341	0.341	0.350	0.344
	taud	2.503	2.502	2.568	2.548	2.577	2.595	2.580	2.540	2.472	2.489	2.493	2.493	2.525
	Ebn at Noon	977	950	934	872	827	808	829	874	911	956	976	976	992
	Edh at Noon	112	108	93	83	70	64	69	82	101	108	113	113	111
All-Sky Solar Radiation	RadAvg	7.12	6.21	4.94	3.43	2.26	1.76	1.97	2.79	3.99	5.35	6.30	6.30	7.04
	RadStd	0.44	0.32	0.28	0.20	0.13	0.14	0.10	0.18	0.28	0.42	0.42	0.36	0.40

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.73	N/A	N/A	+1.17	+0.38	N/A	-18	-142	+225	+94
Regional (37 neighbors)		+0.51	N/A	N/A	+0.84	N/A	N/A	-18	-130	+174	+58

Nomenclature: See separate page